Note that this identity is another version of Theorem 1.7, and it allows one to express the symmetric functions of the e_j's in terms of the Eisenstein series.

2.2 Eisenstein series and divisor functions

We will describe now the link between Eisenstein series and some number-theoretic quantities. This relation comes about if we consider the Fourier coefficients in the Fourier expansion of the periodic function $E_k(\tau)$. Equivalently, we can write $\mathcal{E}(z) = E_k(\tau)$ with $z = e^{2\pi i\tau}$, and investigate the Laurent expansion of $\mathcal{E}$ as a function of z.

We begin with a lemma.

Lemma 2.4 *If* $k \geq 2$ *and* $\mathrm{Im}(\tau) > 0$, *then*

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i\tau\ell}.$$

Proof. This identity follows from applying the Poisson summation formula to $f(z) = 1/(z+\tau)^k$; see Exercise 7 in Chapter 4.

An alternate proof consists of noting that it first suffices to establish the formula for $k = 2$, since the other cases are then obtained by differentiating term by term. To prove this special case, we differentiate the formula for the cotangent derived in Chapter 5

$$\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau.$$

This yields

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

Now use Euler's formula for the sine and the fact that

$$\sum_{r=1}^{\infty} r w^r = \frac{w}{(1-w)^2} \qquad \text{with } w = e^{2\pi i\tau}$$

to obtain the desired result.

As a consequence of this lemma, we can draw a connection between the Eisenstein series, the zeta function, and the divisor functions. The

divisor function $\sigma_\ell(r)$ that arises here is defined as the sum of the ℓ^{th} powers of the divisors of r, that is,

$$\sigma_\ell(r) = \sum_{d|r} d^\ell.$$

Theorem 2.5 *If* $k \geq 4$ *is even, and* $\mathrm{Im}(\tau) > 0$*, then*

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. First observe that $\sigma_{k-1}(r) \leq r r^{k-1} = r^k$. If $\mathrm{Im}(\tau) = t$, then whenever $t \geq t_0$ we have $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, and we see that the series in the theorem is absolutely convergent in any half-plane $t \geq t_0$, by comparison with $\sum_{r=1}^\infty r^k e^{-2\pi r t_0}$. To establish the formula, we use the definition of E_k, that of ζ, the fact that k is even, and the previous lemma (with τ replaced by $m\tau$) to get successively

$$\begin{aligned}
E_k(\tau) &= \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^k} \\
&= \sum_{n\neq 0} \frac{1}{n^k} + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\
&= 2\zeta(k) + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\
&= 2\zeta(k) + 2\sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\
&= 2\zeta(k) + 2\sum_{m>0} \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i m\tau\ell} \\
&= 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \tau m\ell} \\
&= 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.
\end{aligned}$$

This proves the desired formula.

Finally, we turn to the *forbidden* case $k = 2$. The series we have in mind $\sum_{(n,m)\neq(0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but we seek to give it a meaning anyway. We define

$$F(\tau) = \sum_m \left(\sum_n \frac{1}{(n + m\tau)^2} \right)$$

summed in the indicated order with $(n, m) \neq (0, 0)$. The argument given in the above theorem proves that the double sum converges, and in fact has the expected expression.

Corollary 2.6 *The double sum defining F converges in the indicated order. We have*

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r\tau},$$

where $\sigma(r) = \sum_{d|r} d$ is the sum of the divisors of r.

It can be seen that $F(-1/\tau)\tau^{-2}$ does *not* equal $F(\tau)$, and this is the same as saying that the double series for F gives a different value ($\tilde{F}$, the reverse of F) when we sum first in m and then in n. It turns out that nevertheless the **forbidden Eisenstein series** $F(\tau)$ can be used in a crucial way in the proof of the celebrated theorem about representing an integer as the sum of four squares. We turn to these matters in the next chapter.

3 Exercises

1. Suppose that a meromorphic function f has two periods ω_1 and ω_2, with $\omega_2/\omega_1 \in \mathbb{R}$.

(a) Suppose ω_2/ω_1 is rational, say equal to p/q, where p and q are relatively prime integers. Prove that as a result the periodicity assumption is equivalent to the assumption that f is periodic with the simple period $\omega_0 = \frac{1}{q}\omega_1$. [Hint: Since p and q are relatively prime, there exist integers m and n such that $mq + np = 1$ (Corollary 1.3, Chapter 8, Book I).]

(b) If ω_2/ω_1 is irrational, then f is constant. To prove this, use the fact that $\{m - n\tau\}$ is dense in $\mathbb{R}$ whenever τ is irrational and m, n range over the integers.

2. Suppose that $a_1, \ldots, a_r$ and $b_1, \ldots, b_r$ are the zeros and poles, respectively, in the fundamental parallelogram of an elliptic function f. Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r = n\omega_1 + m\omega_2$$